MODEL SELECTION AND SYSTEM ARCHITECTURE

TERRA WATCH

From image changes to evidence-grounded descriptions

Milestone 3 Review

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One image pair. An auditable change description.

Ordered images feed detection and captioning; claims are checked against visual evidence.

01
DETECT
BIT-ResNet18
Road / building mask

→

02
DESCRIBE
RSCaMa
Caption and atomic claims

→

03
INSPECT
Structural witness
Matched regions and crops

→

04
DECIDE
Qwen3-VL
Accept / reject / defer

Input: earlier T1 and later T2 RGB images

Output: accepted description, claim decisions, crop evidence and report JSON

Source masks and reference captions stay in the training/evaluation store.

Model choices follow the task.

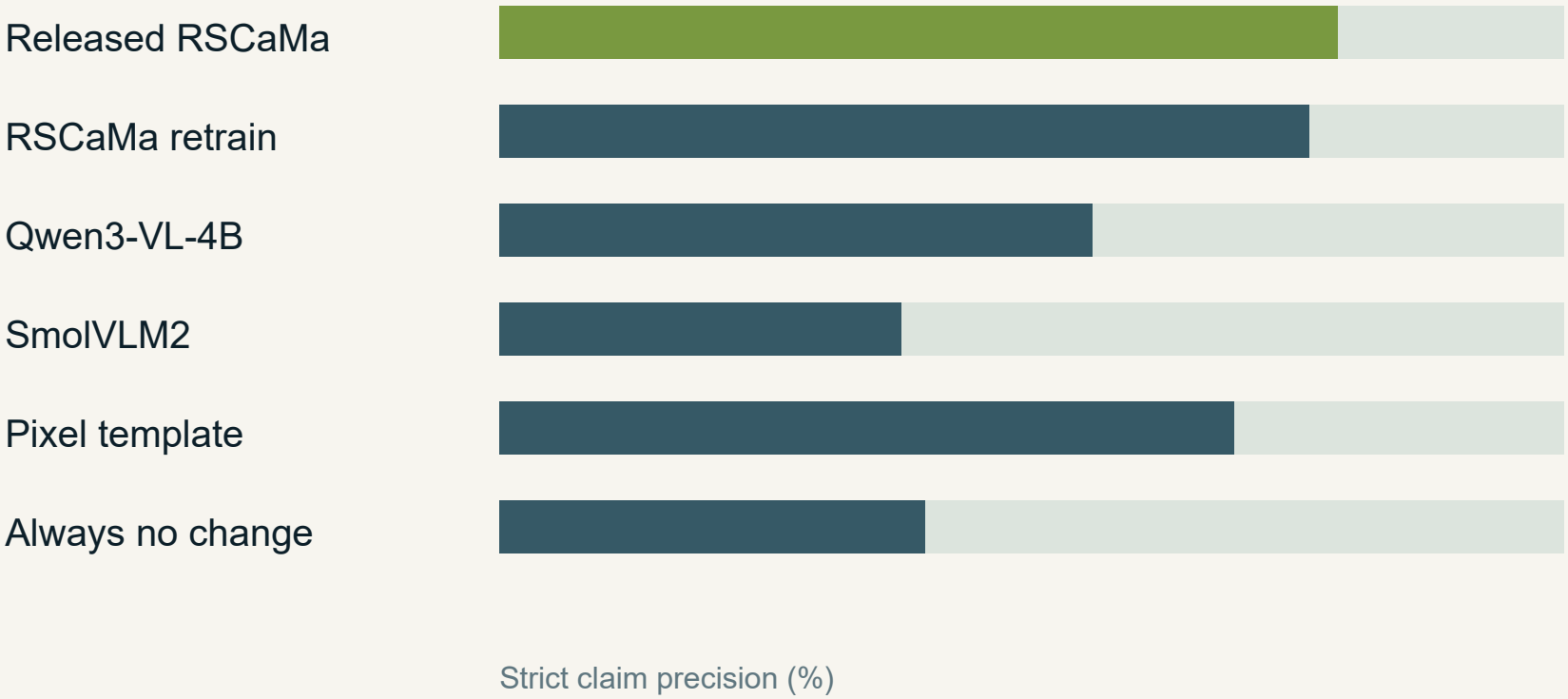
Domain-specific generation combines with dense localisation and explicit claim verification.

BIT-ResNet18	Shared temporal features 256 x 256 inputs; three-class mask
RSCaMa	CLIP + state-space fusion Paired RGB images; caption up to 42 tokens
Visual witness	Parameter-free crop inspection Original-frame crops; quality and change evidence
Qwen3-VL-4B	Image-backed claim judgement Images + claim + evidence; three decisions

Serial model workers isolate dependencies and release GPU memory between stages.

RSCaMa leads the local captioning screen.

100 LEVIR pairs. Automatic claim/reference checks. Same population as the B0 comparison.



0.6223

BLEU-4

0 / 40

unchanged-pair false alarms

Released RSCaMa reproduces the stored B0 aggregates at reported precision.

Localisation and visual evidence support the stack.

Component experiments retain their own samples and metric definitions.

DETECTOR SCREEN

45.8%
BIT binary F1

16.7%
Otsu binary F1

Two epochs: 100 train / 25 validation / 100 evaluation pairs

Unchanged-pair alarms: BIT 31/45, Otsu 45/45

BIT road/building mean IoU: 26.8%

REFEREE SCREEN

56.7%
Image-backed Qwen3-VL agreement

Text / common evidence: 41.3%

VL / common evidence: 45.3%

30 pairs and 150 annotation-proxy claims

Visual witness: 50/50 synthetic checks; 2/3 coarse region checks on authored real-image cases.

Caption source: local rerun summary. Referee source: preserved Colab L4 screen.

Training controls and evidence remain traceable.

The architecture specifies the complete data flow, model interfaces and checkpoint rules.

TRAINING

BIT: ImageNet initialisation; focal + Dice loss

AdamW, cosine decay and validation IoU selection

RSCaMa: CLIP initialisation; token cross-entropy

Adam and development BLEU-4 checkpoint selection

VERIFICATION

83 software checks in the supplied local run

Raw decisions, coordinates and evidence remain attached to each claim

References: M3 report [2-9]; sarthak/evaluation_proof.json; consolidation/evidence/checks.json

REPRODUCIBLE SOURCES

Pipeline interfaces and training:

POC/pipeline/

Captioning and evaluation:

POC/execution/

Local benchmark proof and scripts:

POC/milestone3/sarthak/

Two independent human raters

Cohen's kappa; third-rater adjudication